

Numerical Experiments

Geometry-Dependent Modal Requirements of Finite-Area Interface Representations in Affine-DDPNM

Intended for §6–§10 of the full paper — compiled from the archived benchmark outputs of 2026-08-10

Scope and honesty conventions. This section reports exclusively results that were actually computed by the benchmark pipelines archived under `outputs/benchmark_win` of the three geometry projects (`affine_ddpnm_3d`, `affine_ddpnm_3d_random_porous`, `real_porous_benchmark_3d`). Two reference systems are distinguished throughout (physical FEM reference versus the theoretical broken-pressure C-DD reference); only the former was numerically available, and the distinction is respected in the wording. Items of the planned experimental programme that have *not* been run yet are explicitly listed in §10.3 and are *not* filled in with numbers. Where a speedup is reported, its exact timing scope (first solve vs. online only) is stated.

1 Numerical methodology

1.1 Geometries and discretisations

Three three-dimensional test geometries are used. All are unit-cube packings of solid spheres with Stokes flow in the remaining fluid domain, partitioned into N_p pore subdomains with m planar internal interfaces.

- (i) **Uniform-27**: a regular $3 \times 3 \times 3$ lattice of 27 equal spheres (radius 0.105) centred at $\{0.20, 0.50, 0.80\}^3$. The partition is the regular $4 \times 4 \times 4$ grid cut, giving $N_p = 64$ pores and $m = 144$ interfaces (mean $2m/N_p = 4.5$ ports per pore).
- (ii) **Random-27**: a frozen random packing of 27 spheres (seed 20260804, 18 wall-clipped and 9 interior spheres); the partition is the Voronoi tessellation of the sphere centres, giving $N_p = 27$ pores and $m = 114$ interfaces (mean 8.44 ports per pore).
- (iii) **Real-100**: a frozen random packing of 100 spheres (seed 20260805) with porosity 0.856; the partition is the regular $4 \times 4 \times 4$ grid cut, giving $N_p = 64$ pores and $m = 144$ interfaces (mean 4.5 ports per pore).

The solid-phase packings are shown schematically in Fig. 1 (the finite-element meshes themselves are not rendered; the sphere positions and radii are those used to build the meshes). Mesh statistics are summarised in Table 1, including the objective variability descriptors suggested by the review of the experimental design:

$$\text{CV}(A_\gamma) = \frac{\text{std}(A_\gamma)}{\text{mean}(A_\gamma)}, \quad \text{CV}(V_i) = \frac{\text{std}(V_i)}{\text{mean}(V_i)}, \quad \text{CV}(a_\gamma/b_\gamma) = \frac{\text{std}(a_\gamma/b_\gamma)}{\text{mean}(a_\gamma/b_\gamma)}, \quad (1)$$

where A_γ is the area of interface γ , V_i the volume of pore i , and $a_\gamma/b_\gamma \geq 1$ the in-plane aspect ratio of the interface patch, estimated as the square root of the ratio of the two largest principal-component eigenvalues of the patch vertices. These are *reported* as descriptive quantities; no claim is made that a single scalar controls the error.

The three geometries cover, on the one hand, a deliberately regular case (Uniform-27), a deliberately irregular case (Random-27), and a realistic high-porosity packing (Real-100). Note

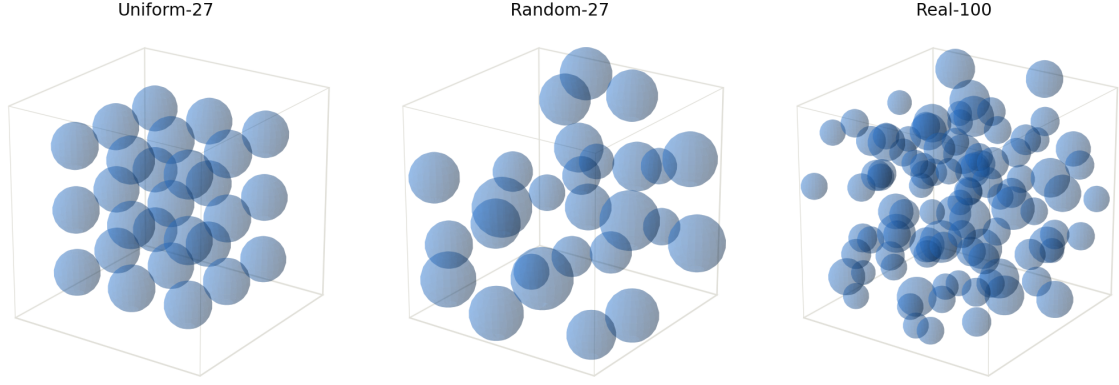


Figure 1: Schematic of the three solid-phase packings used to build the test geometries (sphere positions and radii as used by the mesh generator; finite-element meshes are not shown): (a) Uniform-27 regular $3 \times 3 \times 3$ lattice; (b) Random-27 frozen packing of 27 spheres; (c) Real-100 frozen packing of 100 spheres.

Table 1: Geometry and discretisation summary. FEM DOF counts are for the monolithic Taylor–Hood pair $[P_2]^3-P_1$ (velocity DOFs = $3 \times$ number of P2 nodes; pressure DOFs = number of P1 nodes); they match the mixed-unknown counts reported by the solver.

Geometry	spheres	N_p	m	cells	vel.	pres.	mixed	$2m/N_p$	remarks
Uniform-27	27	64	144	12 203	60 936	3 019	63 955	4.5	regular lattice
Random-27	27	27	114	15 249	72 471	3 483	75 954	8.44	Voronoi, irregular
Real-100	100	64	144	32 611	159 186	7 655	166 841	4.5	grid cut, porosity 0.856

Geometry	CV (V_i)	CV (A_γ)	CV (a_γ/b_γ)	mean a_γ/b_γ
Uniform-27	0.321	0.203	0.158	1.39
Random-27	0.432	0.783	1.479	2.48
Real-100	0.057	0.116	0.078	1.11

that the Real-100 *partition* (regular grid) has small variability descriptors; the irregularity of the *geometry* is carried by the sphere positions and by the pressure coupling, not by the grid partition itself.

1.2 Reference systems and error metrics

Two reference systems must be kept distinct (see also the theory document):

- (i) **Physical/numerical reference:** the monolithic Taylor–Hood finite-element solution $\mathbf{u}_h^{\text{FEM}}, p_h^{\text{FEM}}$ on the *same* mesh, with continuous pressure across pores. All reported accuracy metrics are relative to this reference. The theoretical section does not establish equivalence between this system and the broken-pressure C-DD reference; the error relative to the FEM solution is therefore reported as an *observed* accuracy measure, and no rate statement is made about it from the theory.
- (ii) **Theoretical Galerkin reference:** the broken-pressure C-DD trace solution ξ_{CDD} of the theory sections, and the Schur-energy seminorm $\|\cdot\|_{\hat{\mathbf{g}}_{uu}}$ it induces. The theory-level quantities ($\|\xi_{\text{CDD}} - \xi_X\|_{\hat{\mathbf{g}}_{uu}}$, the Pythagorean decompositions) were *not* computed in the runs archived so far (the full C-DD solve was not executed); they are discussed in §7.3 as planned verification, and no numerical values are reported for them.

Table 2: The four modal interface spaces and their global unknown budgets on the three geometries ($m = \text{number of interfaces: } 144/114/144$). The row marked $\dagger$ was *not computed* in the runs archived so far.

Method	space/interface	modes/iface	global unknowns			
			Uniform-27	Random-27	Real-100	
Classic	$\mathcal{W}_{0n}$	1	144	114	144	Per-interface
P0-vector $\dagger$	$\mathcal{W}_{0v}$	3	432	342	432	
NormalLinear	$\mathcal{W}_{1n}$	3	432	342	432	
Affine	$\mathcal{W}_{1v}$	9	1296	1026	1296	

contents: $\mathcal{W}_{0n} = \{1\mathbf{n}\}$; $\mathcal{W}_{0v} = \{1\mathbf{n}, 1\mathbf{t}_1, 1\mathbf{t}_2\}$; $\mathcal{W}_{1n} = \{1\mathbf{n}, s\mathbf{n}, t\mathbf{n}\}$; $\mathcal{W}_{1v} = \{1, s, t\} \otimes \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$. $\dagger$ Constant-vector (P0-vector) space: basis implemented as part of the four-space framework but not run in the archived benchmarks; included for the equal-budget comparison planned in §8.4.

For each reduced space $X \in \{\mathcal{W}_{0n}, \mathcal{W}_{0v}, \mathcal{W}_{1n}, \mathcal{W}_{1v}\}$ we report, uniformly:

$$e_u^{L^2} = \frac{\|\mathbf{u}_X - \mathbf{u}_{\text{FEM}}\|_{L^2(\Omega)}}{\|\mathbf{u}_{\text{FEM}}\|_{L^2(\Omega)}}, \quad (2)$$

$$e_u^{H_{\text{br}}^1} = \frac{\left(\sum_i \|\nabla(\mathbf{u}_X - \mathbf{u}_{\text{FEM}})\|_{L^2(\Omega_i)}^2\right)^{1/2}}{\left(\sum_i \|\nabla \mathbf{u}_{\text{FEM}}\|_{L^2(\Omega_i)}^2\right)^{1/2}}, \quad (3)$$

$$e_p^{L^2} = \frac{\|p_X - p_{\text{FEM}}\|_{L^2(\Omega)}}{\|p_{\text{FEM}}\|_{L^2(\Omega)}}, \quad (4)$$

$$e_Q = \frac{|Q_X - Q_{\text{FEM}}|}{|Q_{\text{FEM}}|}, \quad Q = \text{outlet volume flux.} \quad (5)$$

All error norms are evaluated cell-wise on the common mesh (broken domain), with degree-6 quadrature. The pressure gauge is fixed by Dirichlet pressure conditions at inlet ($p = 1$) and outlet ($p = 0$) in both the FEM and the reduced systems; the raw relative L^2 pressure error (4) is reported without mean alignment. For the Random-27 geometry an additional mean-aligned variant (the raw error with the global mean pressure difference removed) is available and is reported there; the theory gives no monotonicity of (4) along the nested chains, so occasional non-monotone pressure errors are not a discrepancy.

1.3 Four modal interface spaces

The numerical programme is organised around the four trace subspaces of the theory section, whose per-interface contents are

$$\mathcal{W}_{0n} : \{1 \cdot \mathbf{n}\}, \quad \mathcal{W}_{0v} : \{1 \cdot \mathbf{n}, 1 \cdot \mathbf{t}_1, 1 \cdot \mathbf{t}_2\}, \quad (6)$$

$$\mathcal{W}_{1n} : \{1 \cdot \mathbf{n}, s \cdot \mathbf{n}, t \cdot \mathbf{n}\}, \quad (7)$$

$$\mathcal{W}_{1v} : \{1, s, t\} \otimes \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}, \quad (8)$$

where (s, t) are local planar coordinates on the (representative) interface and $\{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$ a fixed orthonormal frame. Table 2 lists the unknown budgets. The two three-mode spaces satisfy $\dim \mathcal{W}_{0v} = \dim \mathcal{W}_{1n} = 3m$, which makes them an *equal-budget* pair: the same number of interface unknowns spent on directional content ($\mathcal{W}_{0v}$) versus on first-order spatial content ($\mathcal{W}_{1n}$). Of the two paths $\mathcal{W}_{0n} \subseteq \mathcal{W}_{0v} \subseteq \mathcal{W}_{1v}$ and $\mathcal{W}_{0n} \subseteq \mathcal{W}_{1n} \subseteq \mathcal{W}_{1v}$, only the second was executable with the implemented bases at the time of the archived runs; $\mathcal{W}_{0v}$ (constant tangential modes) is listed as *not computed* in this study.

1.4 Timing and solver protocol

All runs used the same serial pipeline on one machine (Windows 11, x86-64; the fenicsx environment; viscosity 1, inlet/outlet pressure 1/0). Timings distinguish

$$T_{\text{offline}} = \text{local factorisations and primitive response construction}, \quad (9)$$

$$T_{\text{online}} = \text{global reduced assembly} + \text{Schur solve} + \text{local reconstruction}. \quad (10)$$

$T_{\text{first}} = T_{\text{offline}} + T_{\text{online}}$ is the wall time of a first reduced solve. The FEM comparison time is the monolithic assembly+solve. Common mesh generation and post-processing validation are excluded from both. Where a “speedup vs FEM” is quoted, the scope is stated: for Uniform-27 and Random-27 it is $T_{\text{first}}(\text{FEM})/T_{\text{first}}(X)$ (a genuine wall-clock speedup including the offline library); for Real-100 the offline cost dominates a single solve and only the online-phase ratio is quoted, together with the absolute times.

2 Verification of the reference implementation

2.1 Exact-FE-Schur: Schur elimination code validation

On Random-27 the benchmark additionally solves the exact Schur complement of the monolithic system on the full FE interface trace (21 970 trace DOFs, 465.9 s). This serves as a code-level validation of the Schur-elimination machinery, *not* as the theoretical C-DD reference: the two systems share the assembled monolithic operator and must agree to round-off. The measured relative difference between the exact-FE-Schur solution and the monolithic solve is

$$\frac{\|\mathbf{u}_{\text{Schur}} - \mathbf{u}_{\text{direct}}\|}{\|\mathbf{u}_{\text{direct}}\|} = 1.05 \times 10^{-12}, \quad (11)$$

with Schur symmetry defect 3.2×10^{-15} , Schur relative residual 5.75×10^{-14} and global relative residual 1.55×10^{-15} . The reduced spaces therefore face a validated reference: their errors relative to the exact Schur solution (column “errors to exact FE-Schur” in the archived report) equal their errors relative to the monolithic solve to 12 digits, confirming that no implementation error is introduced by the reduced-space coupling itself.

2.2 Conservation and algebraic diagnostics

Table 3 collects the algebraic diagnostics recorded by the runs. Local mass conservation is enforced by construction (the interface P0 flux equations are assembled exactly); the residuals shown are the maximum absolute momentum-residual entries of the assembled Schur systems, at round-off level in all cases. The Schur matrices are exactly symmetric by construction (symmetry defect zero for the reduced systems). On Random-27 the minimum Schur eigenvalue drops from 2.1×10^{-8} (Classic) to 1.26×10^{-12} for both NormalLinear and Affine: the three-mode and nine-mode Schur systems are SPD but very ill-conditioned on this geometry; this is a conditioning observation only, no claim about solvability is implied beyond the successful solves.

2.3 Theory-level verification: not yet computed

The theory sections derive the best-approximation property in the $\widehat{\mathbf{S}}^{uu}$ -seminorm and the exact Pythagorean decompositions along the two nested paths, with the reduction errors $\Delta_T = \|\xi_{0v} - \xi_{0n}\|_{\widehat{\mathbf{S}}^{uu}}^2$, $\Delta_N = \|\xi_{1n} - \xi_{0n}\|_{\widehat{\mathbf{S}}^{uu}}^2$, and the identity

$$\|\xi_{\text{CDD}} - \xi_{0n}\|_{\widehat{\mathbf{S}}^{uu}}^2 = \|\xi_{\text{CDD}} - \xi_{1v}\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\xi_{1v} - \xi_{1n}\|_{\widehat{\mathbf{S}}^{uu}}^2 + \|\xi_{1n} - \xi_{0n}\|_{\widehat{\mathbf{S}}^{uu}}^2, \quad (12)$$

and its analogue along $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{0v} \rightarrow \mathcal{W}_{1v}$. These identities are exact for nested spaces in exact arithmetic and their numerical verification requires solving the *full* C-DD system (the broken-pressure Galerkin solution ξ_{CDD}), which was not executed in the archived runs. This verification

Table 3: Numerical verification diagnostics as recorded by the archived runs (“–” = not recorded by that benchmark). “ $\max r_{\text{mom}}$ ” is the maximum absolute momentum residual of the assembled reduced system; the mass-balance row is the inlet/outlet flux sum of the random benchmark.

Geometry	Method	$\max r_{\text{mom}} $	Schur sym. defect	$\lambda_{\min}(\mathbf{S})$	in/out flux sum	
Uniform-27	all	–	–	$1.67 \times 10^{-4}\dagger$	–	
Random-27	W_{0n}	2.17×10^{-16}	0	2.11×10^{-8}	-5×10^{-16}	† Recorded
Random-27	W_{1n}	3.55×10^{-16}	0	1.26×10^{-12}	-2×10^{-16}	
Random-27	W_{1v}	4.92×10^{-16}	0	1.26×10^{-12}	-4×10^{-16}	
Real-100	W_{0n}	6.36×10^{-17}	0	–	–	
Real-100	W_{1n}	1.05×10^{-16}	0	–	–	
Real-100	W_{1v}	1.45×10^{-16}	0	–	–	

by the base-project verification run (`ddpnm_3d_uniform_spheres`, same geometry, Classic system); the affine benchmark did not record the eigenvalue. The monolithic FEM reference solves reached relative linear residuals 1.73×10^{-12} (Uniform-27) and 2.92×10^{-12} (Random-27), with relative mass imbalance 2.54×10^{-14} (Random-27).

Table 4: Complete accuracy results relative to the monolithic Taylor–Hood FEM on the same mesh. $e_u^{L^2}$, $e_u^{H^1_{\text{br}}}$, $e_p^{L^2}$, e_Q as in (2)–(5). †: mean-aligned pressure variant (Random-27 only). The W_{0v} row was not computed in the archived runs.

Geometry	Method	$e_u^{L^2}$	$e_u^{H^1_{\text{br}}}$	$e_p^{L^2}$	$e_p^{L^2, \text{align}\dagger}$	e_Q
Uniform-27	W_{0n}	0.2108	0.4483	0.0321	–	0.1646
Uniform-27	W_{1n}	0.0816	0.2385	0.0185	–	0.0243
Uniform-27	W_{1v}	0.0563	0.2161	0.0217	–	0.0032
Random-27	W_{0n}	0.6532	0.8585	0.1030	0.2196	0.7272
Random-27	W_{1n}	0.3075	0.3897	0.0322	0.0694	0.1465
Random-27	W_{1v}	0.0651	0.1756	0.0203	0.0436	0.0271
Real-100	W_{0n}	0.9254	1.0627	0.0776	–	1.1199
Real-100	W_{1n}	0.4738	0.5129	0.0299	–	0.2595
Real-100	W_{1v}	0.1998	0.3594	0.0236	–	0.1244

is a planned item (see §10.3); the nestedness of the implemented bases is however exact by construction, and the diagnostics of §7.1–§7.2 confirm that the assembled reduced operators are consistent with the monolithic system to machine precision.

3 Controlled modal experiments

3.1 Absolute accuracy

Table 4 collects the complete accuracy results of all archived runs: 3 geometries $\times$ 3 computed spaces, four metrics each. It is an absolute-results table (not improvement ratios).

Three observations follow directly from the absolute numbers, all qualitative and confined to the tested configurations:

- (C1). Constant-normal coupling is insufficient: on every geometry the Classic (W_{0n}) velocity error is at least 21%, and grows to 65%–93% as the geometry departs from the regular lattice.
- (C2). The first-order normal modes W_{1n} remove the dominant part of that error on all three geometries ($e_u^{L^2}$ drops by factors 2.6–2.1–2.0 for Uniform/Random/Real).
- (C3). The additional vectorial content W_{1v} remains significant for Random-27 ($e_u^{L^2}$: $0.308 \rightarrow 0.065$) and Real-100 ($0.474 \rightarrow 0.200$), while on Uniform-27 it is a smaller correction ($0.082 \rightarrow 0.056$). The flux error e_Q on Uniform-27 drops to 0.32% only with W_{1v} .

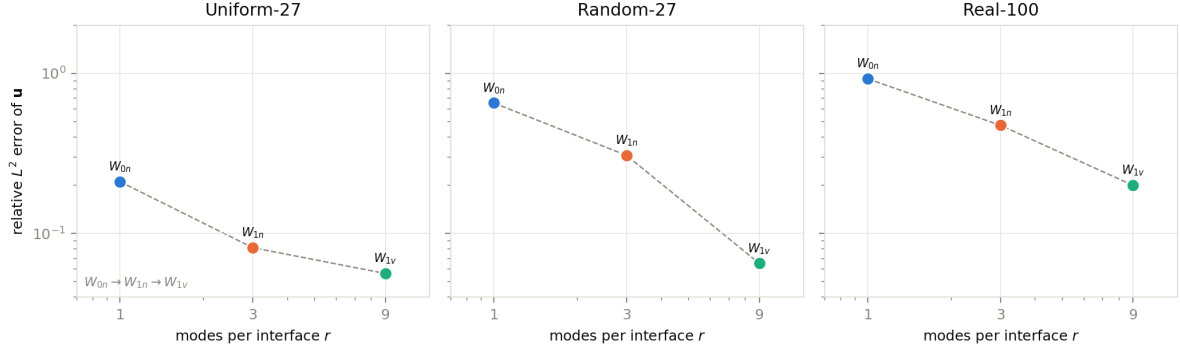


Figure 2: Relative L^2 velocity error versus modes per interface ($r = 1, 3, 9$) along the nested path $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$, for the three geometries (log scale). Only the path through $\mathcal{W}_{1n}$ was computed; the equal-budget space $\mathcal{W}_{0v}$ at $r = 3$ is planned work.

The non-monotonicity of the pressure error on Uniform-27 ($e_p^{L^2}(\mathcal{W}_{1n}) = 0.0185 < e_p^{L^2}(\mathcal{W}_{1v}) = 0.0217$) is not a discrepancy: the theory does not guarantee monotonicity of the pressure L^2 error along nested chains (the reduction error is monotone in the Schur-energy seminorm, not in $L^2(p)$).

3.2 Accuracy versus modal budget

Fig. 2 is the central visual of the section: for each geometry it plots the relative L^2 velocity error against the number of modes per interface, $r \in \{1, 3, 9\}$, joining the points along the computed nested path $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$. The $r = 3$ location carries a single computed space ($\mathcal{W}_{1n}$); the equal-budget companion $\mathcal{W}_{0v}$ is not yet computed, so the question “same DOF budget spent on direction versus on spatial variation” can only be posed here, not answered (see §8.4).

The diminishing-returns shape ($1 \rightarrow 3$ much larger than $3 \rightarrow 9$) is consistent across all three geometries, while the *level* of the remaining error at $r = 9$ differs strongly: 5.6%, 6.5%, 20%. This visual supports the claim that the modal budget requirement is geometry-dependent, without over-interpreting the exact values (no asymptotics in r are claimed).

3.3 Metric comparison across geometries

Fig. 3 shows all four metrics as dot plots (velocity L^2 , broken H^1 , pressure L^2 , outlet-flux error) with logarithmic ordinate; the wide dynamic range (0.3%–112%) is the reason for the log scale. The figure reproduces the Table 3 numbers in a form where the cross-geometry pattern is visible at a glance: the Classic-to-Affine spread widens from the regular to the irregular and realistic cases in every metric except the pressure error, which stays within one decade across all geometries and methods.

3.4 Equal-budget comparison $\mathcal{W}_{0v}$ vs. $\mathcal{W}_{1n}$: planned

The central controlled contrast of the programme is the equal-budget comparison

$$G_{0v} = \frac{e_{0n} - e_{0v}}{e_{0n}}, \quad G_{1n} = \frac{e_{0n} - e_{1n}}{e_{0n}}, \quad (13)$$

i.e. the FEM-relative L^2 reduction obtained by spending three DOFs per interface on constant directional content ($\mathcal{W}_{0v}$) versus on first-order spatial content ($\mathcal{W}_{1n}$). This experiment was *not* run in the archived benchmarks (the constant-vector basis is implemented in the four-space framework but no runs were executed). It is the first item of §10.3. Until it is computed, the statement that the normal-linear modes are the dominant mechanism rests on the comparison $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ alone (Fig. 2), which cannot separate directional from spatial information.

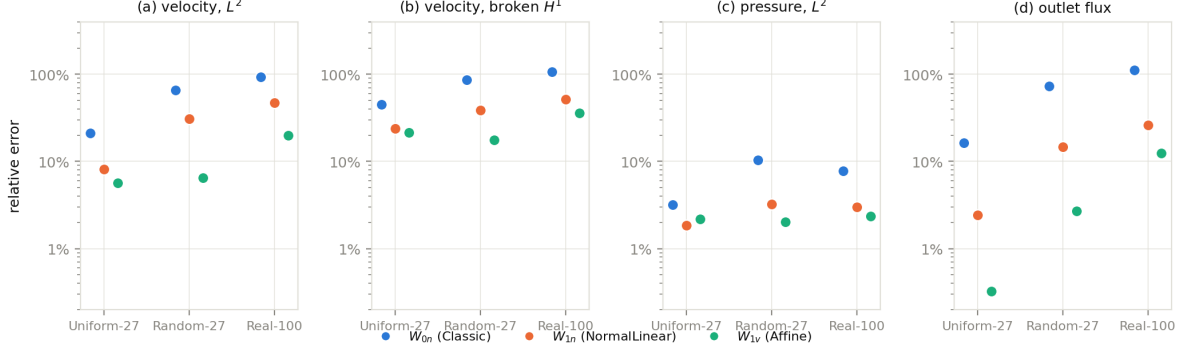


Figure 3: Four error metrics (relative, log scale) across the three geometries, three computed spaces: (a) velocity L^2 , (b) velocity broken H^1 , (c) pressure L^2 , (d) outlet-flux error.

3.5 Field visualization

Fig. 4 shows, for Uniform-27 and Random-27, the mid-plane slice $z = 0.5$ of the FEM velocity magnitude together with the pointwise velocity-error magnitude of the three reduced spaces. Within each row the three error panels share one color scale, so the reduced spaces are directly comparable panel to panel; the reference panel has its own scale. Axes are the physical coordinates of the unit cube. Solid outlines mark the sections of the solid spheres, and grey lines the intersections of the partition interfaces with the slice plane. Along the sequence $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ the error becomes both weaker and more spatially localized. On Uniform-27 the normal-linear space $\mathcal{W}_{1n}$ already recovers most of the structure, while on Random-27 the residual $\mathcal{W}_{1n}$ error is still spread over extended regions and only the affine space $\mathcal{W}_{1v}$ approaches the FEM field, consistent with the global metrics (Table 4).

The slice fields are available for the two controlled geometries; per-interface normal-flux fields and Heterogeneous-100 field outputs remain planned work (see §5.3).

4 Realistic porous structure (Real-100)

4.1 Accuracy

The Real-100 geometry is the largest and most realistic of the three (32 611 cells, 166 841 mixed DOFs, porosity 0.856). Its accuracy rows in Table 4 repeat the qualitative pattern of the controlled geometries, at a higher level: Classic $\mathcal{W}_{0n}$ misses 92.5% of the velocity L^2 norm and even overshoots in the broken H^1 seminorm ($e_u^{H^1} = 1.063$); NormalLinear reduces the L^2 error to 47.4% and the flux error to 25.9%; Affine reaches $e_u^{L^2} = 20.0\%$, $e_Q = 12.4\%$. The pressure error is the least sensitive metric here as well (7.8% $\rightarrow$ 3.0% $\rightarrow$ 2.4%). These are the least accurate of the three geometries for every space, which is consistent with the larger fluid volume and the more convoluted interface network, but no single descriptor is claimed to explain the ordering.

4.2 Cost and memory

Table 5 reports the cost structure of all runs. For Uniform-27 and Random-27 the reduced methods achieve a genuine first-solve speedup over the monolithic FEM ($10.5\times$ and $13.3\times$ for Classic; $8.2\times$ and $9.1\times$ for Affine), because the offline library is cheap relative to the FEM solve on those meshes. For Real-100 the situation is reversed: the monolithic solve takes 14.5s while the reduced offline phase takes 28–33s, so a single reduced solve is *slower* in wall time than the FEM solve; only the online phase is fast (0.06–1.4s, i.e. $255\times$, $72\times$, $10\times$ faster than the FEM solve). The table therefore states absolute times for both phases and both references, and the speedup discussion is confined to the stated scopes.

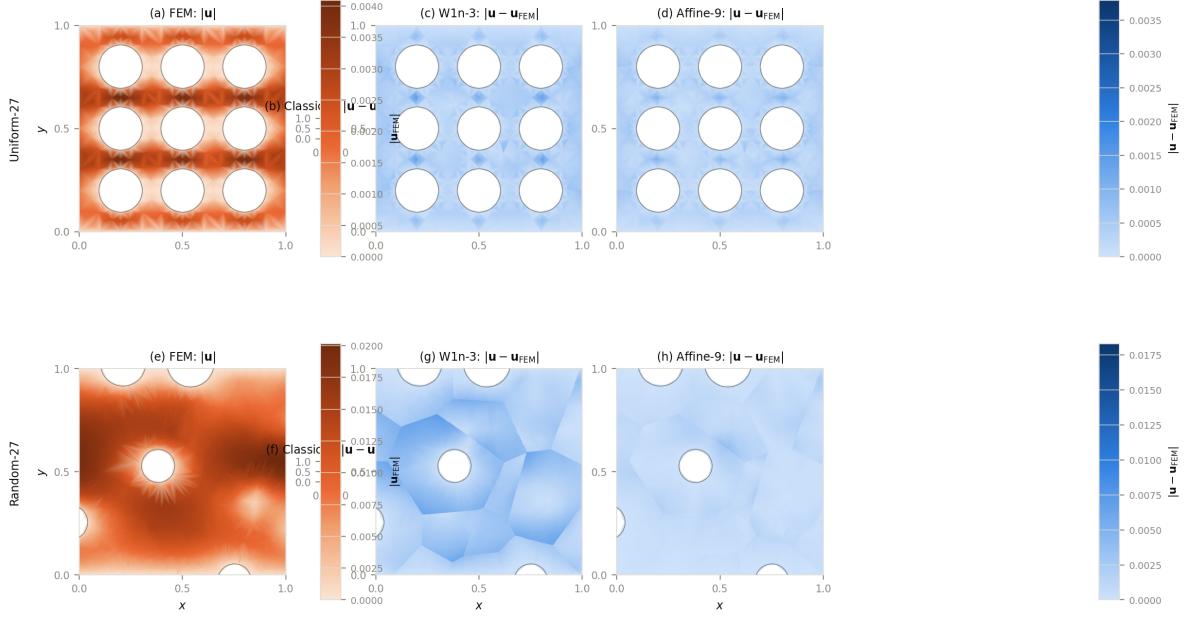


Figure 4: Slice plane $z = 0.5$ of the velocity field and the FEM-relative velocity error on Uniform-27 (top row) and Random-27 (bottom row). Panels (a,e) show the FEM velocity magnitude $|\mathbf{u}_{\text{FEM}}|$; panels (b,f), (c,g) and (d,h) show $|\mathbf{u} - \mathbf{u}_{\text{FEM}}|$ for $\mathcal{W}_{0n}$, $\mathcal{W}_{1n}$ and $\mathcal{W}_{1v}$. Coordinates are physical (unit cube). Within each row the three error panels use a common color scale to enable direct visual comparison; the reference panel has its own scale. Solid outlines are the sections of the solid spheres and grey lines the partition-interface intersections with the slice plane.

Memory: peak footprints on Real-100 are 45 MiB (Classic), 79 MiB (NormalLinear), 246 MiB (Affine) versus 300 MiB for the monolithic solve; all fits comfortably in memory, and the growth with the mode count is consistent with the $9m$ coupling block of the Affine Schur matrix.

4.3 Accuracy–cost tradeoff

Fig. 5 plots, per geometry, the relative L^2 velocity error against the online solve time on log–log axes, together with the monolithic FEM reference point. It answers the question “what accuracy does the added modal information buy” in cost terms. For all three geometries the reduced points lie on a clear accuracy–cost frontier above and to the left of the FEM point; the frontier is strongly geometry-dependent (the Real-100 points at $r \geq 3$ remain above the Uniform-27 $r = 1$ error). The figure does *not* include the offline cost, which is the dominant part of T_{first} on Real-100; it is shown because the online phase is the asymptotically recurring cost in a many-query scenario, and the caption says so.

5 Cross-geometry modal analysis

5.1 Observed FEM-relative error-reduction shares

Along the computed path $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ the observed FEM-relative L^2 velocity-error reduction decomposes into the normal-spatial step ($\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n}$) and the additional vectorial step ($\mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$):

$$R_N = \frac{e_{0n} - e_{1n}}{e_{0n} - e_{1v}}, \quad R_{V|N} = \frac{e_{1n} - e_{1v}}{e_{0n} - e_{1v}}, \quad R_N + R_{V|N} = 1. \quad (14)$$

Fig. 6 and Table 6 report these shares.

Table 5: Computational cost (seconds). T_{offline} : local factorisations + primitive responses; T_{online} : reduced assembly + Schur solve + reconstruction; $T_{\text{first}} = T_{\text{offline}} + T_{\text{online}}$. FEM column = monolithic assembly+solve. N_{prim} = primitive RHS columns. For Real-100, T_{first} is the archived total; FEM there is the monolithic solve only.

Geometry	Method	unknowns	N_{prim}	T_{offline}	T_{online}	T_{first}	speedup vs FEM
Uniform-27	W_{0n}	144	320	7.35	0.0071	7.36	$10.50\times$
Uniform-27	W_{1n}	432	896	7.38	0.0222	7.40	$10.44\times$
Uniform-27	W_{1v}	1296	2624	9.18	0.208	9.39	$8.23\times$
Uniform-27	FEM	63 955	–	–	77.25	77.25	$1\times$
Random-27	W_{0n}	114	246	8.90	0.0141	8.91	$13.29\times$
Random-27	W_{1n}	342	702	9.20	0.0287	9.22	$12.84\times$
Random-27	W_{1v}	1026	2070	12.82	0.191	13.01	$9.10\times$
Random-27	FEM	75 954	–	–	118.42	118.42	$1\times$
Random-27	Exact-FE-Schur	21 970	–	–	465.90	465.90	$0.25\times$
Real-100	W_{0n}	144	–	27.9	0.057	27.98	$255\times$ (online only)
Real-100	W_{1n}	432	–	28.0	0.203	28.17	$72\times$ (online only)
Real-100	W_{1v}	1296	–	32.7	1.400	34.13	$10\times$ (online only)
Real-100	FEM	166 841	–	–	14.53	14.53	$1\times$

Peak memory (Real-100 only): W_{0n} 45.0, W_{1n} 78.8, W_{1v} 246.4, FEM 299.8 MiB. “Online only” speedups are

$T_{\text{online}}(\text{FEM})/T_{\text{online}}(X)$; the absolute times show that a single Real-100 reduced solve is slower than the FEM solve in wall time.

Table 6: Cross-geometry modal summary: absolute errors and the observed FEM-relative reduction shares of (14).

Geometry	$e_{0n}^{L^2}$	$e_{1n}^{L^2}$	$e_{1v}^{L^2}$	R_N	$R_{V N}$	reduction $\frac{e_{0n} - e_{1v}}{e_{0n}}$
Uniform-27	0.2108	0.0816	0.0563	83.6%	16.4%	73.3%
Random-27	0.6532	0.3075	0.0651	58.8%	41.2%	90.0%
Real-100	0.9254	0.4738	0.1998	62.2%	37.8%	78.4%

Two statements can be made from these numbers, both confined to the tested configurations and to the chosen path:

- (D1). On all three geometries the normal-spatial step $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n}$ accounts for the majority of the observed reduction (59–84%).
- (D2). The share of the vectorial step $\mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ is markedly larger on the irregular/realistic geometries (41% and 38%) than on the regular lattice (16%): the relative value of the additional directional content increases with geometric irregularity as measured by the variability descriptors of Table 1 (Random-27 has the largest CV (A_γ) and CV (a_γ/b_γ)). This is an observed association, not a proven causal relation.

5.2 Schur-energy pathwise decomposition: planned

The theory sections give an *exact* decomposition of the squared Schur-energy error into pathwise marginal gains (see §7.3). Unlike the FEM-relative shares of Fig. 6, those quantities (Δ_N , $\Delta_{T|N}$ etc. in the $\hat{\mathbf{S}}^{uu}$ -seminorm) are theorem-level and require the full C-DD trace solution ξ_{CDD} . They were not computed in the archived runs and are therefore not reported numerically; Fig. 6 and the planned Schur-energy figure must not be conflated. The former is an observed descriptive attribution, the latter would be a strict theorem quantity; only the former is available at this stage.

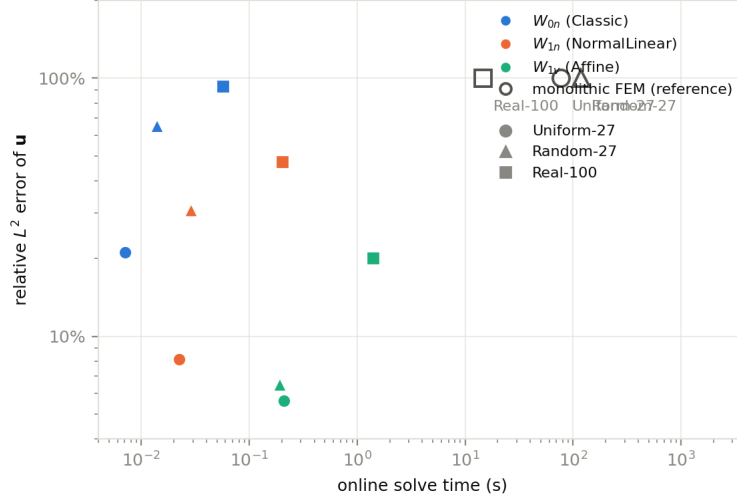


Figure 5: Relative L^2 velocity error versus online solve time (log–log). Marker shape = geometry; colour = space (blue $\mathcal{W}_{0n}$, orange $\mathcal{W}_{1n}$, aqua $\mathcal{W}_{1v}$); hollow markers = monolithic FEM reference. Online time excludes the offline library phase (see Table 5).

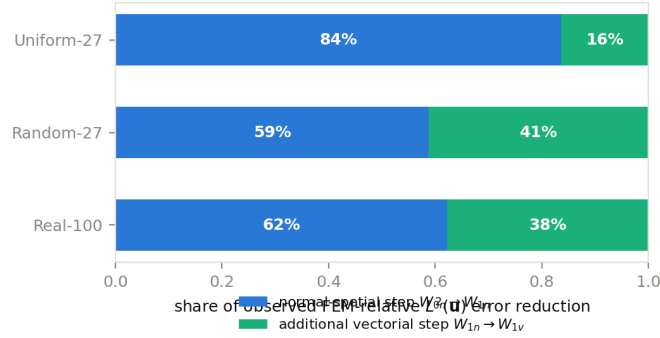


Figure 6: Fractions of the *observed* FEM-relative L^2 velocity error reduction along the prescribed path $\mathcal{W}_{0n} \rightarrow \mathcal{W}_{1n} \rightarrow \mathcal{W}_{1v}$ (100% stacked). These fractions are descriptive; they are not unique causal modal contributions, because the four middle spaces are generally non-nested and the path choice fixes the attribution.

5.3 Geometry dependence and limitations

What the current data supports. (i) constant-normal coupling $\mathcal{W}_{0n}$ is insufficient to represent the finite-area pore interfaces accurately on any of the three geometries; (ii) the first-order normal modes $\mathcal{W}_{1n}$ provide a major correction on all three; (iii) the residual gap between $\mathcal{W}_{1n}$ and $\mathcal{W}_{1v}$ is small on the regular geometry and substantial on the irregular/realistic ones, so the effective low-dimensional modal requirement of an interface *depends on the geometry across the tested configurations*; (iv) no monotonic dependence of the error on a single scalar geometry descriptor is claimed.

What is planned and not yet computed. (a) the equal-budget space $\mathcal{W}_{0v}$ (constant tangential modes) on all three geometries, enabling the controlled contrast of §8.4; (b) the full C-DD solve ξ_{CDD} on Uniform-27 and Random-27 for the exact Schur-energy verification (§7.3) and the pathwise decomposition (§10.2); (c) a mesh-sensitivity study (Uniform-27 at three mesh levels) to confirm that the separation $\mathcal{W}_{0n} \gg \mathcal{W}_{1n} > \mathcal{W}_{1v}$ is not an artifact of one mesh; the theory gives fixed-mesh separation only, no mesh-independent asymptotic statement; (d) an ensemble of random realisations of the Random-27 geometry to test the stability of the 59/41 split across seeds; (e) slice-field outputs for Real-100 once the `z.slice` interface mismatch is fixed; (f) a

second real geometry at a different sphere count.

Scope notes. All timings are single-machine, single-threaded measurements from the archived runs; they are indicative, not benchmark-suite grade. The reported errors are with respect to the monolithic Taylor–Hood FEM on the same mesh; the relation between the broken-pressure C-DD discretisation and the monolithic system is not established in the theory sections, and therefore no claim about the accuracy of the reduced methods *relative to the exact Stokes solution* is made.

Appendix A: archived output provenance

All numbers in this section were produced by the archived benchmark runs of 2026-08-10 and are stored in

- `affine_ddpnm_3d/outputs/benchmark_w1n/`:
`affine_ddpnm_metrics.csv`, `affine_ddpnm_report.json`, `affine_benchmark_fields.npz`;
- `affine_ddpnm_3d_random_porous/outputs/benchmark_w1n/`:
`random_affine_metrics.csv`, `random_affine_report.json`, `random_benchmark_fields.npz`;
- `real_porous_benchmark_3d/outputs/benchmark_w1n/`:
`benchmark_metrics.csv`, `benchmark_report.json`;
- the base-project verification run
`ddpnm_3d_uniform_spheres/outputs/verify_run/` (Classic Schur eigenvalue footnote).

Geometry descriptors were recomputed by the enclosed script (`scripts/geometry_stats.py`) directly from the saved partition meshes (Uniform-27, Random-27) and by rebuilding the grid partition with the benchmark’s own builder (Real-100); the resulting `data/geometry_stats.json` accompanies this section. All figures were regenerated from the CSVs/JSON above by `scripts/make_figures.py`; the field panels of Fig. 4 additionally use the two field archives listed above; no numbers were edited by hand beyond rounding to the significant digits shown.

Appendix B: archived benchmark plot (Uniform-27)

The benchmark pipeline itself produced the four-method comparison plot shown in Fig. 7 (errors, speedup and offline time for the Uniform-27 geometry), which we reproduce here as the archival artifact it is; its bar-chart form is kept exactly as generated, and the colour of the NormalLinear series (`#7B1FA2`) follows the original run, not the palette of the new figures.

One interface entity: $\{1, s, t\} \otimes \{\mathbf{n}, \mathbf{t}_1, \mathbf{t}_2\}$

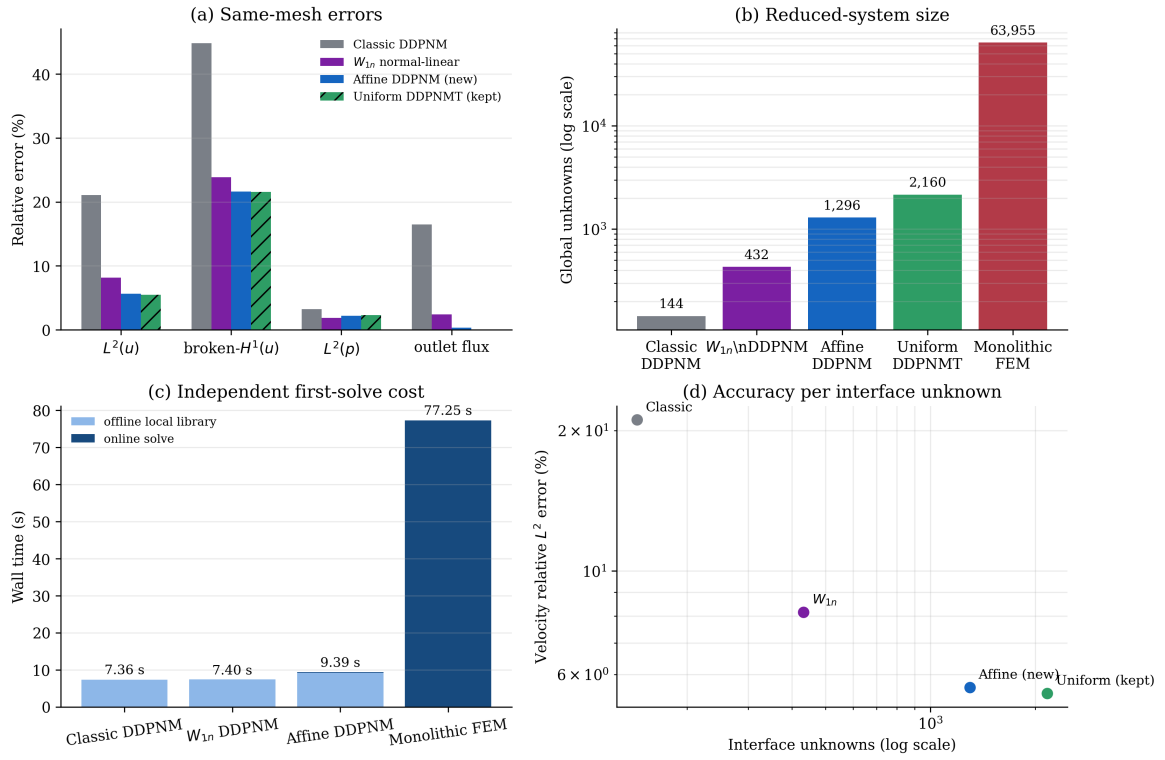


Figure 7: Archived comparison plot generated by the Uniform-27 benchmark run itself (Classic/ W_{1n} /Affine vs. FEM): relative errors (velocity L^2 , broken H^1 , pressure L^2 , flux), speedup and offline time. Reproduced unchanged.